

Savitzky-Golay split command-lag search

rotor median publish period [s]: 0.00499773025513
gimbal median publish period [s]: 0.00499868392944
initial lags [s]: {'rotor': 0.004997730255126953, 'gimbal': 0.004998683929443359}
smooth half-width multipliers: [4.0, 2.0, 1.0, 0.5]

smooth width=4: rotor=0.190106794s gimbal=0.0366704117s cost=1.50142117886
lag_gradient={'rotor': -4.9714040363108e-06, 'gimbal': -6.773777984636631e-07}
FD=[{'direction': 'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 22.895242192372887, 'finite_difference_l2_norm': 22.89524217800757,
'difference_l2_norm': 2.3605773703244653e-07, 'relative_error': 1.0310340246633629e-08},
{'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 8.019175814978167, 'finite_difference_l2_norm': 8.019175770692566,
'difference_l2_norm': 2.5842282533738713e-07, 'relative_error': 3.222560912740017e-08},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 17.157593914710304, 'finite_difference_l2_norm': 17.157593809364187,
'difference_l2_norm': 6.867880132240867e-07, 'relative_error': 4.0028224041091174e-08}]
smooth width=2: rotor=0.188415443s gimbal=0.0363779426s cost=1.50275271575
lag_gradient={'rotor': 2.8239449867406563e-07, 'gimbal': 4.977249947024642e-06}
FD=[{'direction': 'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 28.24400592383532, 'finite_difference_l2_norm': 28.244005231387504,
'difference_l2_norm': 4.264939565855972e-06, 'relative_error': 1.5100335190967931e-07},
{'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 8.784197368646838, 'finite_difference_l2_norm': 8.784196649634431,
'difference_l2_norm': 2.080117266946784e-06, 'relative_error': 2.3680220054837132e-07},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 18.84535245239061, 'finite_difference_l2_norm': 18.845351319121132,
'difference_l2_norm': 4.562611219522373e-06, 'relative_error': 2.42108033322751e-07}]
smooth width=1: rotor=0.190927956s gimbal=0.037239376s cost=1.50291518134 lag_gradient={'rotor':
1.940939539521458e-06, 'gimbal': -4.671503831904511e-06} FD=[{'direction':
'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 21.725257449965234, 'finite_difference_l2_norm': 21.725251574903314,
'difference_l2_norm': 2.1530559936453678e-05, 'relative_error': 9.910381953373875e-07},
{'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 10.633314231507544, 'finite_difference_l2_norm': 10.633308706255333,
'difference_l2_norm': 1.2494860858479153e-05, 'relative_error': 1.1750673953992316e-06},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 21.832166020640372, 'finite_difference_l2_norm': 21.832157926237613,
'difference_l2_norm': 2.7473455612421847e-05, 'relative_error': 1.2583934908908322e-06}]
smooth width=0.5: rotor=0.190958181s gimbal=0.0378565992s cost=1.50238376892
lag_gradient={'rotor': -2.1128445313184363e-06, 'gimbal': -3.717842513684122e-06}
FD=[{'direction': 'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 23.495016938605254, 'finite_difference_l2_norm': 23.494978930562684,
'difference_l2_norm': 9.907860329242535e-05, 'relative_error': 4.217004973919674e-06},
{'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 13.922687985616143, 'finite_difference_l2_norm': 13.922648432199463,
'difference_l2_norm': 8.783587535682555e-05, 'relative_error': 6.308830266653311e-06},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 24.650500118714547, 'finite_difference_l2_norm': 24.65043924623026,
'difference_l2_norm': 0.00015078823312627187, 'relative_error': 6.1170455933993055e-06}]
strict profile iterations: 1
iteration 1: best=(0.190958180968, 0.037856599215) cost=1.50317435046
rotor_range=(0.185960450713, 0.195955911223) gimbal_range=(0.032857915286, 0.042855283145)

expansions=[]

selected rotor lag 0.190958180968 s (38.209 publish periods)
selected gimbal lag 0.037856599215 s (7.57331 publish periods)
selected objective cost 1.50317423028